CLASS 8 · CBSE · GANITA PRAKASH PART 2 · DPP SHEET

# Chapter 4: Exploring Some Geometric Themes

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. How many squares remain at Step 3 of the Sierpinski Carpet, and how many holes are there?
2. What fraction of the area remains at Step 2 of the Sierpinski Carpet?
3. How many sides does the Koch Snowflake have at Step 3?
4. Find the faces, edges and vertices of a hexagonal prism.
5. Find the faces, edges and vertices of a pentagonal pyramid.
6. Name a solid whose profile is a circle from one direction and a triangle from another.
7. A cylinder has radius 3.5 cm and height 12 cm. Find the sides of the rectangle in its net. (Take $\pi = \frac{22}{7}$.)
8. A cuboid is 4 cm by 3 cm by 5 cm (length, breadth, height). An ant goes from a bottom corner to the opposite top corner along the surface. Find the shortest distance.
9. What are the front, top and side views of an upright cylinder?
10. True or false: the projection of a line segment can be longer than the segment.

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## Answer Key

1. 512 squares and $1 + 8 + 64 = 73$ holes
2. $\frac{64}{81}$
3. $3 \times 4^3 = 192$
4. 8, 18 and 12
5. 6, 10 and 6
6. A cone
7. 12 cm and 22 cm
8. The smallest of $\sqrt{74}$, $\sqrt{90}$, $\sqrt{80}$, that is $\sqrt{74}$ cm, about 8.6 cm
9. Rectangle, circle, rectangle
10. False. It is equal to or shorter than the segment.

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